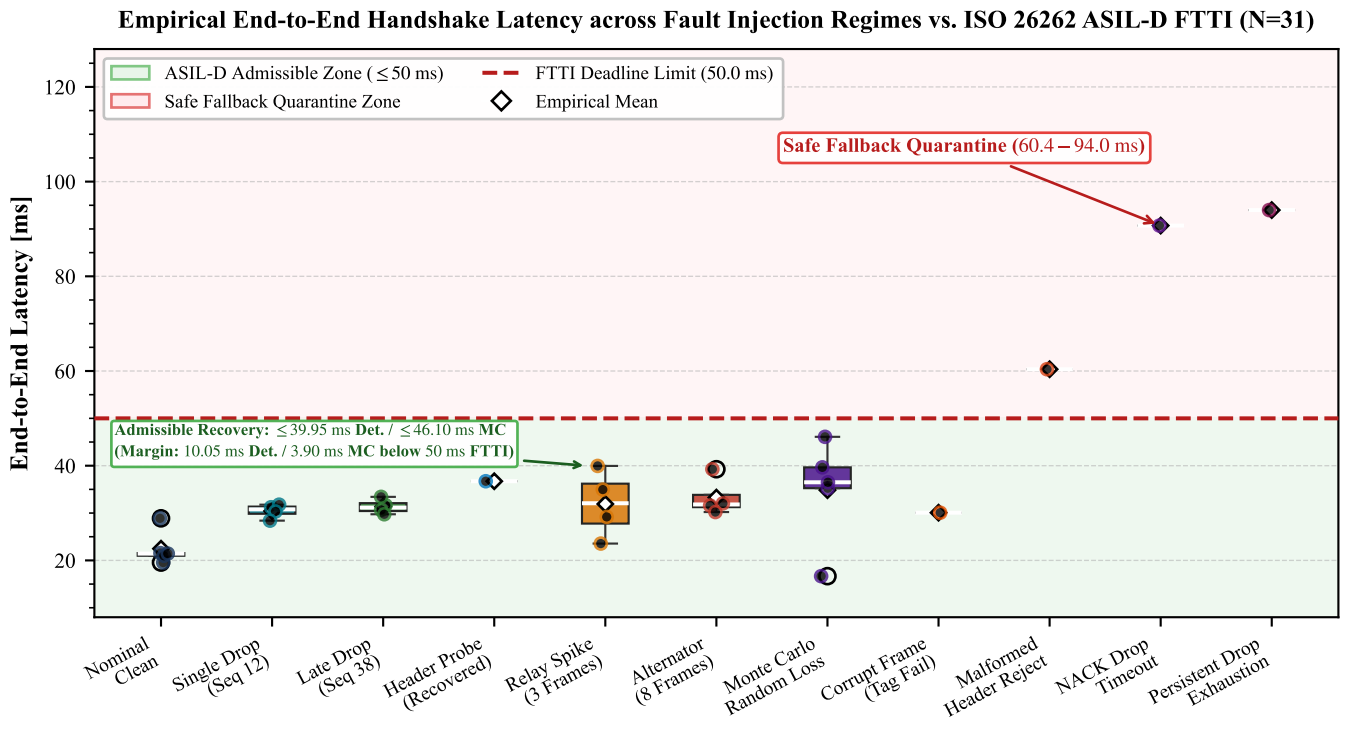

Empirical End-to-End Handshake Latency across Fault Injection Regimes vs. ISO 26262 ASIL-D FTTI (N=31)
ASIL-D Admissible Zone (≤ 50 ms)
Safe Fallback Quarantine Zone
FTTI Deadline Limit (50.0 ms)
Empirical Mean
Safe Fallback Quarantine (60.4 – 94.0 ms)
Admissible Recovery: ≤ 39.95 ms Det. / ≤ 46.10 ms MC
(Margin: 10.05 ms Det. / 3.90 ms MC below 50 ms FTTI)
End-to-End Latency [ms]
120
100
80
60
40
20
Nominal Clean
Single Drop (Seq 12)
Late Drop (Seq 38)
Header Probe (Recovered)
Relay Spike (3 Frames)
Alternator (8 Frames)
Monte Carlo Random Loss
Corrupt Frame (Tag Fail)
Malformed Header Reject
NACK Drop Timeout
Persistent Drop Exhaustion